

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. The genuine past kernel of the prepared Sinai process contracts in a weighted Wasserstein metric; remote deterministic tails are discounted rather than falsely minorized in total variation. An exponential multiplier algebra makes the Feynman–Kac and Doob operators well defined, the Poisson martingale is exactly centered, and a uniform quenched enhanced invariance principle follows from coupling and bracket control. The continuous-time resolvent is derived from entrance, excursion, and residual renewal operators. Memory descriptors are extracted only from the final Schur complement after pole–zero cancellation, and integrable vertical bounds produce a canonical instantaneous, descriptor, and exponentially decaying residual decomposition.

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1. ROUND-FOURTEEN POSITIVE CLOSURE: WEIGHTED COUPLING, ROUGH KERNELS, AND SCHUR-COMPLEMENT MEMORY

The state is the genuine symbolic past and is metrized so that a remote deterministic tail is exponentially discounted. We use weighted Wasserstein contraction, not a common total-variation minorization. Feynman–Kac potentials belong to an exponential multiplier algebra, and memory descriptors are extracted from the final Schur complement after all pole–zero cancellations.

1.1. The past kernel. Let $\mathbf{H}$ be the graph-completed past space of A3. For $0 < \beta < 1$, put

$$d_\beta(h, \tilde{h}) = \sum_{j \geq 0} \beta^j (1 \wedge d_E(h_{-j}, \tilde{h}_{-j})), \quad W(h) = 1 + \sum_{j \geq 0} \beta^j W_E(h_{-j}).$$

The normalized Gibbs kernel $P(h, dh')$ appends one excursion and shifts the past.

Theorem 1.1 (Weighted Wasserstein spectral theorem). *There are $m \geq 1$, $\rho < 1$, and $C < \infty$ such that*

$$\mathcal{W}_{d_\beta}(P^m(h, \cdot), P^m(\tilde{h}, \cdot)) \leq \rho d_\beta(h, \tilde{h}), \quad P^m W \leq \rho W + C.$$

Consequently P has a unique invariant probability π , and on

$$\|f\|_{\beta, W} = \sup_h \frac{|f(h)|}{W(h)} + \sup_{h \neq \tilde{h}} \frac{|f(h) - f(\tilde{h})|}{d_\beta(h, \tilde{h})(W(h) + W(\tilde{h}))}$$

the centered operator has an exponential spectral gap.

Proof. The normalized one-step g -function has summable past variations: if two pasts agree in their last r excursions, bounded distortion gives

$$\|P(h, \cdot) - P(\tilde{h}, \cdot)\|_{\text{TV}} \leq C\vartheta^r.$$

Use maximal coupling until the first common new excursion. Once coupled, the newly appended coordinates agree and the old discrepancy is shifted one place, hence multiplied by β . If coupling fails, the distance is bounded by the weighted tail. Choosing m so that $C\vartheta^m + \beta^m < 1$ gives the contraction.

The exponential excursion moment from A3 and the fact that a shift multiplies the remote contribution by β give $P^m W \leq \rho W + C$. The weak Harris argument can be proved directly: iterate the coupling for the Lipschitz seminorm and use the drift estimate for the weighted supremum norm. The resulting Doeblin–Fortet inequality and compactness of weighted sublevels give the spectral gap. $\square$

1.2. The admissible Feynman–Kac algebra. For $\eta > 0$, define

$$\mathcal{V}_\eta = \{V : \|V\|_{\text{Lip}} < \infty, \quad |V(h)| \leq c_V + \eta \log W(h)\}.$$

Choose η below the exponential moment in the drift theorem.

Lemma 1.2 (Exponential multiplier chart). *There is $\eta_0 > 0$ such that, for $V \in \mathcal{V}_{\eta_0}$ in a neighborhood of zero,*

$$P_V f = P(e^V f)$$

is a bounded analytic family on a slightly heavier weighted Lipschitz space. It has a simple positive eigenvalue λ_V , positive eigenfunction h_V , and dual eigenmeasure π_V , all analytic in V .

Proof. Since $e^{|V|} \leq e^{c_V} W^{\eta_0}$, Hölder's inequality and the exponential excursion moment give $P(e^{|V|} W) < \infty$ uniformly on the source ball. The product Lipschitz estimate follows from $|e^V - e^{\tilde{V}}| \leq e^{\max(V, \tilde{V})} |V - \tilde{V}|$. Thus the exponential power series converges in operator norm on a heavier weight. The gap in Theorem 1.1 and analytic Riesz perturbation give the eigen-data. $\square$

Define the Doob kernel

$$P_V^D f(h) = \frac{1}{\lambda_V h_V(h)} P(e^V h_V f)(h).$$

Theorem 1.3 (Doob nonlinear tower). *The kernel P_V^D is Markov and has the same weighted Wasserstein spectral properties on a smaller chart. The nonlinear operators*

$$\mathcal{E}_n^V F = \log(P_V^D)^n(e^F)$$

satisfy $\mathcal{E}_{n+m}^V = \mathcal{E}_n^V \mathcal{E}_m^V$.

Proof. The eigenfunction identity gives $P_V^D 1 = 1$. Positivity is immediate. Analytic closeness to P preserves the coupling and drift inequalities. The tower follows from the linear semigroup law before taking the logarithm. $\square$

1.3. Centered Poisson decomposition and a quenched rough limit.

For centered $g \in \text{Lip}_{\beta, W}$, set

$$h_g = \sum_{j \geq 0} P^j g, \quad D_{k+1} = h_g(H_{k+1}) - P h_g(H_k).$$

Then D_{k+1} is a martingale difference and

$$\sum_{j=1}^n g(H_j) = \sum_{j=0}^{n-1} D_{j+1} + P h_g(H_0) - P h_g(H_n).$$

Theorem 1.4 (Uniform quenched enhanced invariance principle). *For every finite vector of centered weighted Lipschitz observables, the rescaled additive process, its roof interpolation, and its second iterated integrals converge, uniformly for initial pasts in a weighted compact set, to a Brownian rough path*

$$(B_t, \mathbb{B}_{s,t}), \quad \mathbb{B}_{s,t} = \int_s^t (B_r - B_s) \otimes \circ dB_r + (t-s)\Gamma.$$

The covariance and area anomaly are the absolutely convergent Green–Kubo and antisymmetric correlation sums.

Proof. The spectral gap makes the Poisson series converge in the weighted Lipschitz norm. Conditional centering follows from $\mathbb{E}[D_{k+1} \mid \mathcal{F}_k] = Ph_g(H_k) - Ph_g(H_k) = 0$. The coupling estimate gives uniform convergence of predictable brackets on weighted compact sets, while the exponential moment gives the conditional Lindeberg bound. The martingale functional CLT follows.

Apply the same decomposition to tensor products and use the Burkholder inequality to obtain tightness of the iterated integrals. The antisymmetric expectation of one-step cross terms converges to Γ . A3 supplies the joint renewal law of the roof and terminal residual; its exponential moment and deterministic-clock LDP imply uniform negligibility of the interpolation remainder. The coboundary is $o(\sqrt{n})$ uniformly on weighted compact sets by the drift bound. $\square$

1.4. Continuous-time renewal resolvent. Let $\mathcal{U}_t^V$ be the suspension Doob semigroup and L_V its generator on $L^2(\Pi_V)$. For a finite-dimensional resolved subspace $\mathcal{R} \subset D(L_V^2)$, let $P_{\mathcal{R}}$ be the orthogonal projection and

$$C_V(z) = P_{\mathcal{R}}(z - L_V)^{-1}P_{\mathcal{R}}.$$

Entrance, full-excursion, and terminal-residual operators from A3 give the renewal identity

$$(z - L_V)^{-1} = E_V(z) + A_V(z)(I - \mathcal{L}_{V,z})^{-1}B_V(z)$$

on the common strip furnished by A2.

Theorem 1.5 (Renewal-resolvent vertical theorem). *There is $\gamma > 0$ such that the displayed identity continues meromorphically to $\Re z > -\gamma$. After subtracting its finite Laurent principal parts in any compact substrip, the first two vertical derivatives satisfy*

$$\|\partial_z^j R_{\text{reg}}(\sigma + ib)\| \leq C(1 + |b|)^{-1-\delta-j}, \quad j = 0, 1, 2,$$

uniformly for $-\gamma/2 \leq \sigma \leq 1$.

Proof. Laplace transformation of the first-return decomposition gives the renewal identity, including the entrance and residual operators. On bounded frequencies, analytic Fredholm theory applied to the A2 parent bundle gives the meromorphic continuation. On large frequencies, the integrable four-range estimate of A2 applies to $(I - \mathcal{L}_{V,z})^{-1}$; the entrance and residual operators gain one roof derivative, and two integrations by parts give the stated vertical powers. Compactness leaves only finitely many poles in a compact substrip, counted with algebraic multiplicity. $\square$

1.5. Memory after pole-zero cancellation. Define the final Schur complement

$$\widehat{K}_V(z) = zP_{\mathcal{R}} - P_{\mathcal{R}}L_V P_{\mathcal{R}} - C_V(z)^{-1}.$$

All descriptor data are extracted from $\widehat{K}_V$, not separately from poles of C_V .

Theorem 1.6 (Canonical descriptor–memory decomposition). *The meromorphic matrix $\widehat{K}_V$ has a unique decomposition*

$$\widehat{K}_V(z) = P_{\text{inst}}(z) + \sum_{\zeta \in \mathcal{Z}} \sum_{\ell=1}^{m_\zeta} \frac{A_{\zeta,\ell}}{(z - \zeta)^\ell} + \widehat{K}_{\text{reg}}(z),$$

where P_{inst} is the polynomial part at infinity, the finite set $\mathcal{Z}$ and residues are the actual principal parts of the final Schur complement after every pole–zero cancellation, and

$$\|\partial_z^j \widehat{K}_{\text{reg}}(\sigma + ib)\| \leq C(1 + |b|)^{-1-\delta-j}.$$

The inverse Laplace transform is

$$K = P_{\text{inst}}(\partial_t)\delta_0 + K_{\text{desc}} + K_{\text{reg}}, \quad \|K_{\text{reg}}(t)\| \leq Ce^{-\gamma t/2}.$$

The descriptor realization is minimal after quotienting unreachable and unobservable modes, and is unique up to similarity.

Proof. Compute the Smith–McMillan form of the finite matrix $\widehat{K}_V$ in the meromorphic strip. Principal parts are taken only after inversion of C_V , so the scalar example $C(z) = (z + 1)^{-1}$ gives $\widehat{K} = 0$ and contributes no descriptor. The large- z expansion on $\mathcal{R} \subset D(L_V^2)$ gives the polynomial instantaneous part. The vertical theorem and the matrix inverse identity preserve the residual decay away from the finitely many genuine zeros. Bromwich contour shift is justified by the integrable vertical bound, producing the exponential estimate. Standard controllable/observable reduction of the finite principal-part realization gives minimality and similarity uniqueness. $\square$

Corollary 1.7 (Exact projected Volterra equation). *For $A \in D(L_V)$, $x(t) = P_{\mathcal{R}}\mathcal{U}_t^V A$ satisfies the causal generalized Langevin equation whose memory distribution is Theorem 1.6; the rough scaling of this equation is driven by the enhanced limit in Theorem 1.4.*

Proof. Laplace transform the projected evolution and solve the Schur-complement identity for $C_V(z)^{-1}$. Uniqueness of causal Laplace transforms gives the equation. Joint convergence follows from the integrable residual kernel, the finite descriptor coordinates, and the rough-path continuous mapping theorem. $\square$

Theorem 1.8 (Round-fourteen A4 closure). *The prepared Sinai history has a proved weighted Wasserstein spectral gap, a correctly centered quenched rough tangent, an admissible Doob nonlinear tower, and a canonical descriptor–memory decomposition based on the final Schur complement.*

Proof. Combine Theorems 1.1, 1.3, 1.4, 1.5, and 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and

build gates are internal reproducibility checks and do not replace independent external mathematical review.

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